Marc Riera CITP

HPC and AI Infrastructure Architect | Infrastructure and Platform Leader | CITP MBCS
Cambridgeshire, UK | marc@riera.co.uk
linkedin.com/in/joanmarcriera/

Profile

Infrastructure architect and hands-on technical lead for HPC, storage and platform services, in infrastructure roles since 2007. At Bull and Atos, from April 2011 to September 2016, I installed and operated HPC clusters for research centres with Slurm and InfiniBand, moved Lustre storage at the Barcelona Supercomputing Center, then worked in managed services, solution design and pre-sales. At EMBL-EBI, from September 2016 to March 2026, I started on the LSF compute farms and became service owner for the archive, transfer and identity platforms. The FIRE archive passed 150PB of active research data. I want to design and deliver HPC and AI systems. I have also led: eight direct reports, about 30 engineers influenced, about £1m of budget and more than 4,500 Linux servers and VMs.

Technical record

- BSC MinoTauro at Bull, 2011 to 2014, my first full cluster installation: NVIDIA Tesla M2090 GPU blades, InfiniBand, water-cooled rear doors, CUDA drivers, Slurm with a highly available controller, acceptance node by node and then across the fabric.
- Lustre at BSC: a 1PB system moved to a new 2PB one, using bonded interfaces on the InfiniBand cards and Slurm-scheduled copy jobs, then the old capacity added to it.
- Atos Tenerife: a cluster of about 700 nodes consolidated to 250 and rebuilt from bare metal with Katello, PXE and SaltStack in about 16 minutes a pass, with OpenStack and Hortonworks Hadoop on top.
- FIRE active archive at EMBL-EBI: from about 15PB at ownership handover to more than 150PB active by March 2026, with about 400PB of raw storage underneath, behind one HTTP upload API and S3-compatible reads.
- Transfer services: 40 mount points reduced to 3 and 80 virtualised Globus installations to 3. Tickets fell from close to 100 a week to a handful a month; effective throughput rose from 10 to 16Gbps to a sustained 60 to 75Gbps.
- Tape DR: I wrote and ran the tender. Staged acceptance tests led to the PoINT Archival Gateway, which writes S3 objects to IBM tape and cut disaster recovery total cost of ownership by 70%.
- EMBL-EBI's first zero-downtime data-centre migration; I advocated the stretched VLANs behind it.

Core skills and technologies

HPC and compute: Slurm, LSF and Sun Grid Engine schedulers; bullx suite; NVIDIA GPU nodes and CUDA drivers on BSC MinoTauro at Bull, 2011 to 2014; Linpack and IOzone acceptance testing; bare-metal provisioning with Katello, PXE and SaltStack.
Storage and transfer: petabyte object storage and tape; S3 APIs; ActiveScale, Dell EMC Isilon and ECS, NetApp, VAST Data, Lustre on DDN, Ceph; PoINT Archival Gateway on IBM tape; Globus and Aspera.
Networking and data centre: InfiniBand fabric design with Voltaire and Mellanox subnet management; Ethernet bonding and LACP; stretched VLANs; 100Gbps JANET/Jisc link; MetalLB in BGP mode; rack layout, cabling, power and cooling planning, commissioning.
Platform, cloud and automation: Kubernetes on bare metal and virtual machines with kubeadm and RKE; OpenStack; VMware in pre-sales design; AWS and GCP cost modelling; configuration management with Puppet, Ansible and SaltStack; Terraform for virtual load balancers; Rundeck; GitLab CI/CD, mandatory code review, unit and functional tests; Bash, Python and awk as daily tools.
Identity and Access Management: Red Hat Identity Management, FreeIPA, OpenLDAP, Active Directory trust; two-factor onboarding; bastion-host privileged access; joiner and leaver processes; HashiCorp Vault.
Delivery and pre-sales: tender writing and staged acceptance tests; solution design and proofs of concept; TCO and cost models; customer training and handover documentation; ITIL 4 incident, request and change practice; ServiceNow; Checkmk, Zabbix, Prometheus and Grafana.

Current work in AI infrastructure

- My production GPU cluster work dates from Bull, 2011 to 2014. Since then GPU work has been study and labs.

- A multi-GPU, multi-node training and benchmark lab on rented GPU cloud nodes in October 2026, and a home lab on TrueNAS with Prometheus and Grafana GPU telemetry on a shared NVIDIA RTX 4060, alongside Ollama and Whisper inference containers.
- NVIDIA Academy courses in September and October 2026, listed under credentials: AI factory reference architectures, InfiniBand administration, Cumulus Linux, SONiC, NetQ and cable validation. Planned exam, November 2026: NVIDIA-Certified Professional, AI Networking.
- Schedulers, storage and networks in production between those dates: three LSF farms of about 300 nodes at EMBL-EBI, the proposal to move from LSF to Slurm, archive growth of about 25PB a year by 2026 and a sustained 60 to 75Gbps of transfer traffic.

Experience

Study and personal time, Apr 2026 to present

Since leaving EMBL-EBI in March 2026 I have taken time for study and for my family: NVIDIA Academy AI infrastructure and networking courses, September and October 2026; Google Professional Cloud Architect, September 2026; Chartered IT Professional status awarded by BCS, August 2026; Oxford Saïd Executive Finance Programme, May 2026.

EMBL-EBI, Hinxton, UK. Sep 2016 to Mar 2026

Service owner for the archive, transfer and identity platforms; Deputy Head of Operations, Jan 2021 to Mar 2025

The deputy title came with covering the Head of Operations' duties for about four years, through the vacancy and the new Head's joining and probation; without the decision power that comes with the Head's post, I managed through influence across a team serving two of the institution's five missions.

- Service owner for archive, transfer, identity and monitoring services, accountable for their reliability, cost and delivery. The estate held more than 4,500 Linux servers and VMs. More than 50,000 researchers used its public data, from over 80 countries.
- We moved EMBL's identity estate from OpenLDAP with bespoke schemas to Red Hat Identity Management with standard schemas and an Active Directory trust, for 5,000+ federated users across six sites. I hired and mentored the team that did it.
- Privileged administrators fell from about 50 to 3, through a bastion host. Two Oracle and two MySQL databases were retired. HR records became the authoritative source: no account without a signed contract, a single-use activation link with two-factor setup for joiners, and accounts closed on the contract end date.
- Advocated stretched VLANs and helped sequence the cutover with networking and service teams. EMBL-EBI completed its first data-centre migration with no service interruption. No addresses changed.
- From 2021, when new hardware arrived, FTP and HTTP moved from two Kubernetes clusters of 10 to 12 machines to three bare-metal hosts. Kubernetes was removed where it did not fit. The workload chose the platform.
- All Transfer Services tickets now route to the Service Desk. I trained its staff ticket by ticket and wrote their documentation. Rundeck lets them run delegated actions without seeing scripts or credentials.
- Eight direct reports, about 30 engineers influenced and an annual infrastructure budget of about £1m. About £520k secured through the SITSC infrastructure roadmap for a large CAD data transfer and processing project in 2025.

Compute, then archive, transfer and identity services lead, Sep 2016 to Dec 2020

- Joined in the compute team: about a year helping run three LSF farms of about 300 nodes, and from day one the 25-node LSF cluster that ran the transfer services as LSF-scheduled containers. I rebuilt it, with self-healing and monitoring scripts that adjusted LSF queue configuration, and proposed on cost grounds the move from LSF to Slurm that the institute made years later.
- FIRE took five archive teams, each with its own staging area, database and pull daemon, and replaced that with one HTTP upload API and S3-compatible reads. Old and new ran together. Reads moved first, then writes, each team switching when ready.
- Java and Spring Boot services on Kubernetes replaced the earlier Python stack, built through GitLab CI/CD with mandatory code review, unit and functional tests, and one image promoted unchanged through every environment. Provisioning in Puppet, Ansible and SaltStack.
- I wrote and ran the tender for an S3-compatible tape tier. Staged acceptance tests led to the PoINT Archival Gateway, deployed in August 2020; it writes S3 objects to IBM tape through a RAM cache, with no disk buffer.

- UK Biobank, 2020: 3.9PB ingested during the COVID-19 response; I proposed Globus over Aspera.

Bull SAS and Atos, Barcelona and Spain. Apr 2011 to Sep 2016

Linux and HPC systems engineer and trainer; technical lead, managed services in Barcelona; Technical Service Manager from 2015; Solutions Architect in the final period. Bull was acquired by Atos in 2014.

- Joined as an L3 Linux engineer and trainer, working on HPC at L2, for the Barcelona Supercomputing Center, CESCA and CSIC. I configured and operated Slurm and worked with LSF and the bullx suite.
- MinoTauro in full: racks and water-cooled rear doors in, blades racked and flashed, Ethernet bonding, InfiniBand and GPU cabling, GPU blade firmware and BIOS issues resolved, scripts around the ib* diagnostics for link errors. I trained the customer's administrators on Slurm and the bullx suite and maintained the cluster afterwards.
- Further installations: PRBB in Barcelona, the gravitational-wave group at the Universitat de les Illes Balears, Centro de Investigación Príncipe Felipe in Valencia, storage and tape at CEFCA in Teruel, and a CPU cluster at CSIC Barcelona.
- I designed the InfiniBand fabrics with Voltaire and Mellanox subnet management, and worked on storage from Lustre on DDN to Ceph up to 5PB and EMC Celerra up to 2PB.
- Acceptance with Linpack for compute and IOzone for storage, results gathered with awk to find the bad cable or the bond without LACP. Rack layout, cabling, power and cooling planning, commissioning. I contributed to tenders for national supercomputing procurements in Spain.
- The Tenerife rebuild was a proof of concept: three complete rebuilds in two weeks. It helped my business unit secure ClickFox service delivery, about EUR 1m in associated internal services revenue.
- Designed and deployed a SaltStack orchestration system for about 4,500 Unix and Windows servers, and set up FreeIPA to federate authentication for about 400 Active Directory help-desk operators.
- Automated OpenStack deployment for FIWARE's distributed infrastructure. ARAG Spain: pre-sales design for 25 Oracle systems, then installation of the development, pre-production and production systems.
- Co-founded Bull Spain's managed-services group in Barcelona with its manager and an ITIL lead. I owned the technical design, the L1 and L2 engineers' operating documentation and mentoring, and L3 escalation.
- Bull's Top Contributor recognition in three consecutive years; Atos Global Talent Pool in 2016.

Barcelona Media, now Eurecat, Barcelona. May 2007 to Mar 2011

Research support coordinator and infrastructure lead

- Ran Linux system administration, identity, messaging, monitoring, backup and grid systems as the organisation grew from about 50 to 250 people, with Puppet and SVN for version-controlled provisioning and Ganglia and Zabbix for monitoring.
- An eight-GPU compute system bought from Bull, which supported Red Hat and its bullx suite; I installed it on Debian with Sun Grid Engine and designed the queuing, running within a week of delivery. Bull hired me on the strength of it.

Professional credentials

AI and GPU infrastructure: NVIDIA Academy, September 2026: AI Factory Reference Architectures; AI Infrastructure and Operations Fundamentals; Introduction to Networking; InfiniBand Essentials; Cumulus Linux Essentials; NetQ Deployment and Installation; MLXLink and MLXCables; SONiC Essentials. October 2026: InfiniBand Network Administration; Cable Validation Tool Fundamentals. NVIDIA Deep Learning Institute, 2025: Fundamentals of Accelerated Data Science; Adding Knowledge to LLMs.
Cloud: Google Professional Cloud Architect, September 2026; Open Telekom Cloud Architect, April 2026.
Earlier technical: Red Hat RHCSA and RHCE, 2011, lapsed; Oracle Database 11g OCP, 2015.
Service management: ITIL 4 Strategist Direct, Plan and Improve, 2026; PRINCE2 Agile Practitioner and Foundation, 2020; EXIN IT Service Management Foundation, ISO/IEC 20000, 2012.
Professional standing: Chartered IT Professional, CITP MBCS, BCS, awarded August 2026.

Education

BSc Computer Science and Engineering, La Salle, Universitat Ramon Llull, Barcelona, 2005. Final-year project at the University of Strathclyde, 2006.

Cambridge Judge Business School, Chief Technology Officer Programme, 2025.

Oxford Saïd Business School, Executive Finance Programme, May 2026.